



Project Catalyst: AtliQo Credit Card Launch

Data-Driven Customer Segmentation & Statistical A/B Validation

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The Gen-Z Credit Gap

Most companies guess what their customers want. We decided to prove it with data.

- Data scale

AtliQo Bank possessed a massive database of 500,000+ transaction records.

- Key blind spot

The 18–25 cohort is 25.9% of the base but shows near-zero credit card adoption.

- Objective

Design a targeted product and mathematically prove ROI for capturing this untapped segment.



Mining the Opportunity

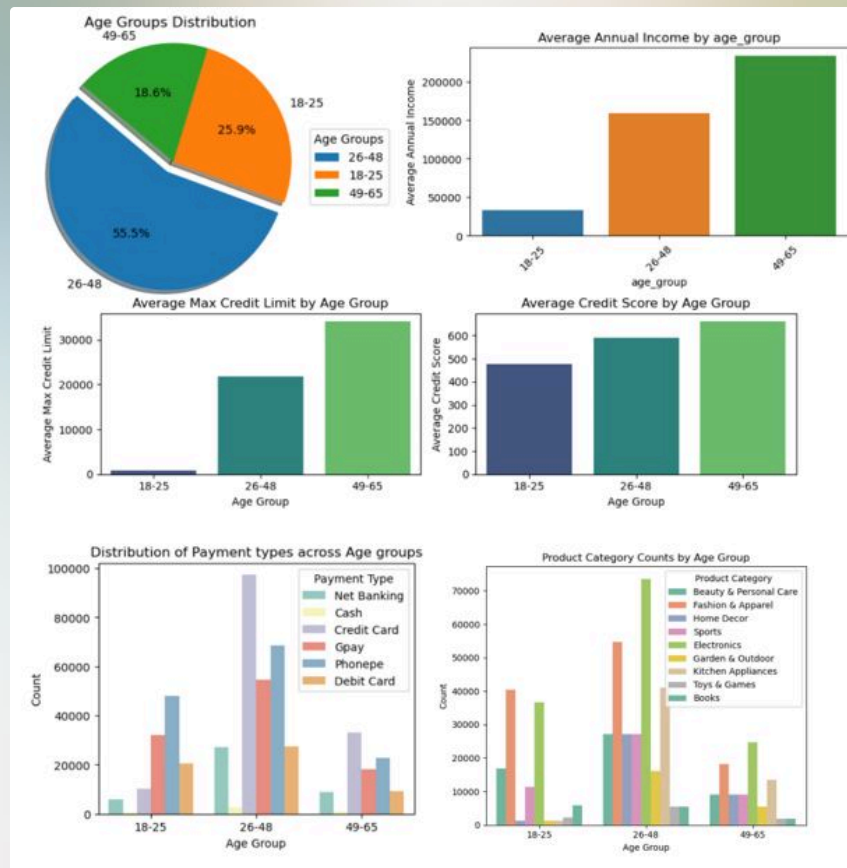
Raw data extraction and rigorous EDA to surface product-market fit signals for 18–25-year-olds.

Pipeline & tooling

- Data extracted from MySQL, processed with Python (Pandas / NumPy).
- Programmatic occupation-based median imputation preserved dataset variance (no row drops).
- Reproducible ETL notebooks and schema versioning enforced.

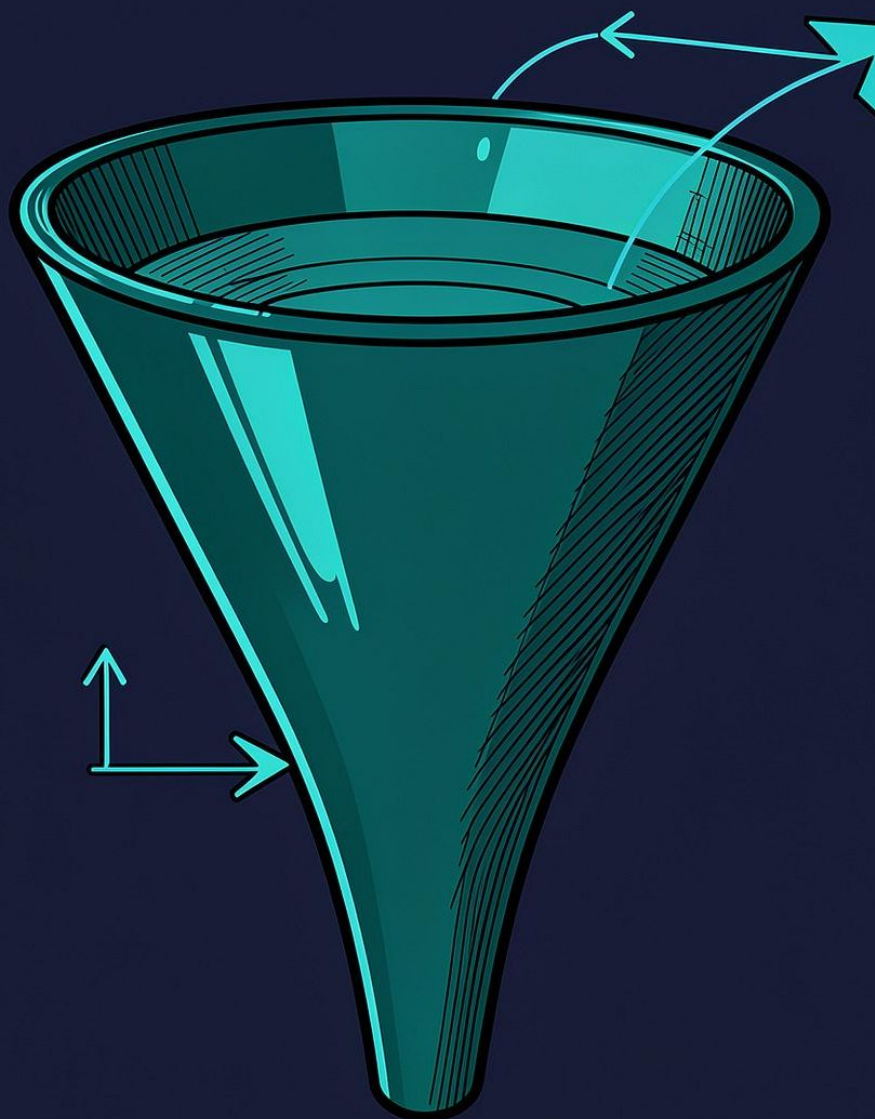
EDA highlights

- 18–25 segment spends heavily in Retail, Electronics, and Beauty.
- Payment behavior: dominant UPI (PhonePe / GPay) and Debit usage—low card penetration.
- Opportunity: convert habitual UPI users to a light-credit product for daily spend.



The Go-To-Market Strategy

How do we issue credit to a generation that shuns traditional cards? Build trust, align with existing payment habits, and minimize friction.



Product

Entry-level, UPI-integrated credit card with micro-credit lines for daily purchases.

Acquisition

In-app offers targeting top UPI spenders, social onboarding, instant virtual card issuance.

Risk & Credit

Alternative underwriting using transaction behavior, rent/utility proxies, and occupation-based priors.

Retention

Gamified credit-building milestones, cashback on preferred categories, education nudges.

❏ Callout: Launch an entry-level, UPI-integrated credit card for 18–25 year olds to build credit via small, daily purchases.

From Intuition to Scientific Proof

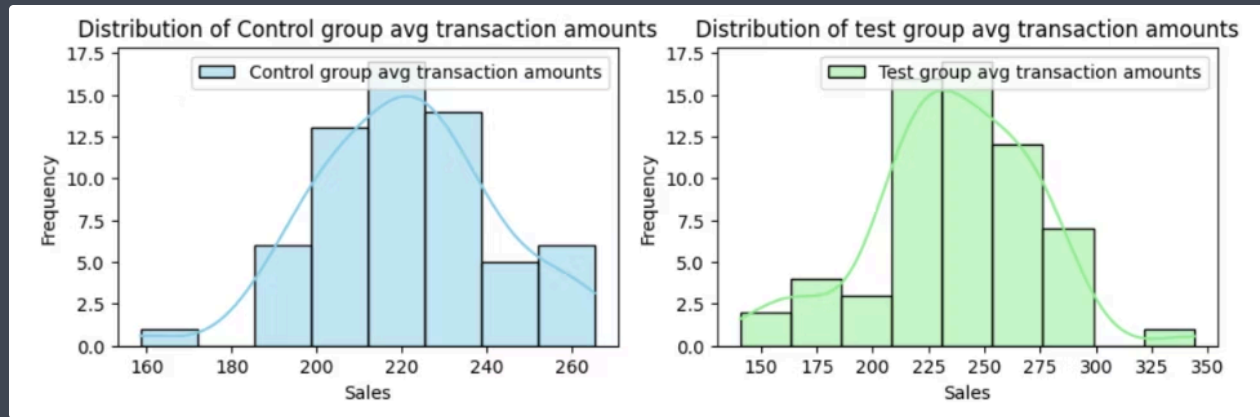
We converted strategy into a controlled experiment: a 62-day pilot with strict isolation between test and control cohorts.

Design & sample

- Sample size calculated with statsmodels: target power 0.8, alpha 0.05, effect size 0.4 $\rightarrow$ $n=100$.
- Randomized assignment: 40 users in test, 40 users in control (isolated), remainder reserved for QA and buffer.

Execution

- 62-day campaign window, daily KPI tracking.
- Reached 40 active conversions in test cohort (40% conversion rate from targeted outreach).
- Isolation ensured no cross-contamination of offers or messaging.



The Verdict: A Statistically Proven Success

Primary KPI: Average Transaction Amount over 62 days. The targeted product generated a measurable uplift.

221.18

Control Group Avg

Baseline average transaction amount

235.98

Test Group Avg

Average transaction amount with new card

+6.69%

Spend Increase

Relative lift in average transaction amount

Statistical result: Right-tailed two-sample Z-test $\rightarrow Z = 2.748$, $p = 0.0029$. With $p < 0.05$ we reject the null hypothesis; uplift is statistically significant.

Data Translates into Business Growth

Business impact

- Captured a previously untapped, high-frequency spending segment (18–25).
- Demonstrated product-market fit with measurable increase in transaction volume.
- Validated that light-credit, UPI-friendly products can monetize habitual mobile payers.

Next steps

- Scale pilot to a statistically powered production launch (N → 10k+), with stratified sampling across metros.
- Refine underwriting models using extended behavioral features and real-time risk scoring.
- Roll out loyalty and credit-building features to increase retention and LTV.

Full Jupyter Notebook, statistical proofs, and methodology available on GitHub.

[agarwal07harsh-ux/atliqo-credit-card-analytics: Analyzing 500k+ banking transactions to uncover untapped demographics and statistically validate a new credit card launch using A/B testing.](#)

